Cyclic Incremental Latin $n \times n$ Squares encoding Generalized NC-Coins Squares for $0 \leq c \leq n$

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Abstract

For every natural number $n \notin \{0, 2, 3, 4, 6\}$, there exists a permutation $q : \mathrm{Fin}\, n \to \mathrm{Fin}\, n$ such that $L_{i,j} = (i + q(j)) \bmod n$ is an *Cyclic Incremental Latin Square*, encoding NC-Coins Squares for $0 \leq c \leq n$.

An NC-Coins Square for $0 \leq 1 \leq n$ corresponds to a solution of n-Queens Problem.

We have an existence proof in `Lean` and construction code in `Haskell`, `Julia` and `Lean`.

See e.g. MO#510135, PSE#138939, MO#516225 and GitHub.

The proof uses the *Injective-Majorization Principle* that, in combination with the *Pigeon Hole Principle*, replaces intensive checking inequalities on each diagonal to the existence of a single function $f(i, j) \leq L_{i,j}$ that is injective on every diagonal.

The proof uses four families covering almost all natural number modulo 6 classes

- for $n \equiv 1, 5 \pmod 6$ (with $n \geq 1$)
- for $n \equiv 2, 4 \pmod 6$ (with $n \geq 10$)
- for $n \equiv 0 \pmod 6$ (with $n \geq 12$)
- for $n \equiv 3 \pmod 6$ (with $n \geq 27$)

For all four families a Cyclic Incremental Latin Square can be constructed. Constructing a Cyclic Incremental Latin Square for the four uncovered natural numbers, $n \in \{8, 9, 15, 21\}$, is handled separately. Likewise, constructing a (non-Cyclic) Incremental Latin Square for the two remaining natural numbers, $n \in \{4, 6\}$, is also handled separately.

1 Cyclic Incremental Latin Squares

1.1 From NC-Coins Squares to Cyclic Incremental Latin Squares

NC-Coins Squares are subsets $S_c \subseteq \{0, \ldots, n-1\}^2$ of nc cells of an $n \times n$ square such that every row, column, sum-diagonal ($i + j = s$), and difference-diagonal ($i - j = d$) contains at most c cells of S_c.

Clearly, for $c = 1$, they correspond to solutions of n-Queens Problems.

The construction of Le, Li and Wang [1] constructs S_c's separately for each c.

1.2 The Conjecture

Every permutation $q : \mathrm{Fin}\, n \to \mathrm{Fin}\, n$ defines a *cyclic Latin square* (a column-permuted Cayley table of $\mathbb{Z}/n\mathbb{Z}$):

$$L_{i,j} = (i + q(j)) \bmod n \qquad (0 \le i, j < n).$$

Because each column is a cyclic shift of $(0, 1, \ldots, n-1)$ by $q(j)$ and q is a bijection, every row and every column of L is automatically a permutation of $\{0, \ldots, n-1\}$.

The conjecture is that for all $n \ge 1$ except $n \in \{2, 3, 4, 6\}$, there exists a column permutation q such that $L_{i,j} = (i + q(j)) \bmod n$ is an incremental Latin square. (For $n \in \{2, 3\}$, even the $k = 0$ layer—the classical n-queens problem—has no solution; for $n \in \{4, 6\}$, exhaustive finite search confirms that no cyclic Latin square satisfies the diagonal bounds.)

Theorem 1.1 (Cyclic Incremental Latin Square Conjecture)**.** *For every integer $n > 0$ with $n \notin \{2, 3, 4, 6\}$, there exists a bijection $q : \mathrm{Fin}\, n \to \mathrm{Fin}\, n$ such that the cyclic Latin square*

$$L_{i,j} = (i + q(j)) \bmod n$$

satisfies:

(i) For every $i \in \mathrm{Fin}\, n$, the row map $j \mapsto L_{i,j}$ is bijective.

(ii) For every $j \in \mathrm{Fin}\, n$, the column map $i \mapsto L_{i,j}$ is bijective.

(iii) For every $0 \le k < n$ and every $s \in \mathbb{N}$,

$$|\{(i, j) \in \mathrm{Fin}\, n \times \mathrm{Fin}\, n \mid i + j = s \ \wedge\ L_{i,j} \le k\}| \le k + 1.$$

(iv) For every $0 \le k < n$ and every $d \in \mathbb{Z}$,

$$|\{(i, j) \in \mathrm{Fin}\, n \times \mathrm{Fin}\, n \mid i - j = d \ \wedge\ L_{i,j} \le k\}| \le k + 1.$$

References

[1] Mao-Hua Le, Wen Li, and Edward T. Wang, *A generalization of the n-queen problem*, Journal of Systems Science and Mathematical Sciences **3** (1990), no. 2, 183–192, https://sysmath.cjoe.ac.cn/jssc/CN/Y1990/V3/I2/183.